Janhavi Pohnerkar

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EDUCATION

B-Tech, Electronics and Telecommunication — **2023-Present**
MIT Academy of Engineering
CGPA- 8/10 (Till SY BTech) | SGPA – 8.6/10

10th | CBSE | Percentage – 90% — **2011- 2021**
PSBA English School, Aurangabad

12th | HSC | Percentage – 69%
Vidyadham Junior College, Aurangabad — **2021- 2023**

WORK EXPERIENCE

R&D Intern | Computer Vision | IOT — **June 2025- Present**
Copper Cloud Industry 4.0 Solutions — ***Pune***

- Currently involved in two **computer vision and image processing projects** focused on quality analysis in the textile and mesh manufacturing industries, one of the projects having **reached deployment stage** in the industry.
- Received comprehensive training on IoT architectures and communication protocols, with emphasis on integration and optimization of connected industrial systems for Industry 4.0 applications.

Projects

- **High Precision Coordinate Measurement System for a Mesh Manufacturing Industry based in USA**
  Developed a computer vision–based solution for defect detection and dimensional measurement in metal mesh manufacturing using **DIPlib** for advanced image processing and **sub-pixel measurement**. The system achieves an accuracy of **300 microns**, identifying irregularities in mesh geometry and dimensions through automated image analysis, thereby strengthening quality assurance and significantly reducing manual inspection time.

- **Color-Based Defect Detection System for Zari Thread Manufacturing (Textile Industry, Surat)**
  Developed a computationally efficient computer vision–based defect detection and color classification system for metallic Zari threads using CIE L*a*b* (LAB) color space **for illumination-invariant color segmentation**. The algorithm identifies surface defects and metallic color variations through deterministic image processing techniques**, eliminating the need for machine learning–based solutions** and significantly reducing computational overhead.

- **Digital Elevation Model (DEM) Generation of Lunar and Martian Surfaces**
  Developed a computer vision–based system for generating high-resolution Digital Elevation Models (DEMs) of lunar and Martian terrains using **photoclinometry (shape-from-shading)** techniques. Implemented **Lambertian and Hapke reflectance models** to accurately estimate surface normals and reconstruct terrain elevation from satellite imagery, enabling detailed topographical and geological analysis of extraterrestrial surfaces.

- **Myoelectric Bionic Arm Prosthesis for Gesture-Based Control**
  Designed and developed a **servo-actuated 3D-designed** myoelectric bionic arm prosthesis for assistive and rehabilitation applications. It is a system capable of **identifying 7 distinct hand gestures** using **EMG (myoelectric) signals**, enabling intuitive prosthetic control. Applied **machine learning–based classification techniques in MATLAB** to map muscle activity patterns to corresponding gesture commands.

TECHNICAL SKILLS

- **Programming**
  C, C++, Python, MATLAB | Python Libraries: OpenCV, DIPlib, NumPy, Pandas, Matplotlib, scikit-learn, Joblib, tensorflow
- **Computer Vision & Image Processing**
  Image Filtering, Edge Detection, Segmentation, Morphological Operations | Object Detection and Geometric Feature Analysis | Sub-pixel Coordinate Measurement and Dimensional Analysis | Photoclinometry / Shape-from-Shading (Lambertian and Hapke Models) | 2D Data Interpretation for Surface and Depth Estimation
- **3D Modeling & Design**
  Fusion 360, AutoCAD, Autodesk Maya
- **Embedded Systems & IoT**
  Microcontrollers: STM32, ESP32, NodeMCU, Raspberry Pi | Communication Protocols: Node-Red | MQTT, UART, I2C, SPI | IoT System Architectures
- **Deployment, Web & Development Tools**
- Web Technologies: HTML, CSS | Flask | Model Serialization & Inference: Joblib | Version Control & Collaboration: Git

SOFT SKILLS

- **Languages-** English (Professional), Hindi (Professional), Marathi (Native),
- During my internship, I strengthened essential soft skills such as **effective communication**, **adaptability**, and **time management**. I collaborated with cross-functional teams to solve complex problems, consistently meeting deadlines and delivering results. This experience helped me develop greater resilience, emotional intelligence, and leadership skills.

COURSES & CERTIFICATIONS

Digital Twin - certificate

Iot Architectural Engineering - L&T- certificate

Python Programming - certificate

Personal Details

Gender- Female | **Date of Birth** – 22 Jan 2005 | **Emails** – Janhavi.pohnerkar@gmail.com, 202301090067@mitaoe.ac.in